CS399P: Major Technical Project | Mid-Term Evaluation

Sim-to-Real Medical Robotics:

Interactive 3D Ultrasound Simulation & Learning Platform

A physics-grounded, hospital-grade simulation environment for autonomous ultrasound scanning and VLA benchmarking

OBJECTIVE 1: SIMULATION PLATFORM
PyBullet 1000 Hz dynamics coupled with 3D Slicer via OpenIGTLink, GPU CT acoustic reslicing, and virtual safety fixtures.

OBJECTIVE 2: LEARNING & BENCHMARKING
Active Next-Best-View planning, multimodal VLA & diffusion policies, and sim-to-real transfer with tissue-mimicking phantoms.

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1. PROBLEM & BACKGROUND

Clinical Motivation: The Robotic Ultrasound Dilemma

Why autonomous ultrasound requires a physics-grounded simulation foundation

OPERATOR DEPENDENT

High Cognitive & Physical Load

• Sonography requires subtle, continuous adjustments of 6-DoF probe orientation, tilt, and acoustic gel coupling.

• Over 80% of sonographers report work-related musculoskeletal disorders (WRMSD) due to sustained contact force application.

• Diagnostic accuracy varies significantly with practitioner experience and fatigue.

COUPLED DYNAMICS

Perception Coupled to Force

• Unlike optical camera robotics, an ultrasound probe sees NOTHING without continuous physical contact with skin.

• Too little force: acoustic acoustic dropouts and massive shadow artifacts.

• Too much force: tissue deformation, patient discomfort, and internal organ displacement.

DATA SCARCITY

Sim-to-Real is Mandatory

• Reinforcement learning or trial-and-error training directly on human patients is ethically and practically impossible.

• Real-world robot teleoperation on physical silicone phantoms is slow, costly, and lacks anatomical diversity.

• A physics-faithful simulator is the ONLY viable pipeline to generate millions of paired multimodal training sweeps.

Core Takeaway: To train AI policies for healthcare, simulation must model contact forces, soft-tissue compliance, and acoustic wave formation simultaneously.

1. PROBLEM & BACKGROUND

Project Objectives: Platform Development & AI Benchmarking

Decomposing the MTP into two synergistic technical milestones

OBJECTIVE 1

Simulation Platform Development

Building a clinical-grade, physics-accurate simulation environment:

• High-Frequency Dynamics: 1000 Hz PyBullet physics simulating a Kinova Gen3 7-DoF arm with realistic joint friction, limits, and soft tissue contact.

• Medical Suite Integration: Direct OpenIGTLink binary link into 3D Slicer at 60 FPS without middleware or ROS overhead.

• GPU Acoustic Wave Physics: PyTorch GPU ray reslicing on patient CT scans modeling acoustic attenuation, impedance reflections, and speckle noise.

• Virtual Safety Fixtures: Enforcing keep-out safety zones and contact force boundaries to guarantee patient safety constraints.

OBJECTIVE 2

Learning & Benchmarking Suite

Developing autonomous AI policies and benchmarking sim-to-real transfer:

• Active 3D Reconstruction: Autonomous volumetric scanning via submodular Next-Best-View (NBV) planning to maximize volume coverage with minimal sweeps.

• Multimodal VLA Policies: Benchmarking Vision-Language-Action models and diffusion action-chunking conditioned on acoustic images, force, and text instructions.

• Sim-to-Real Transferability: Validating policy transfer from virtual patient CTs to physical silicone phantoms and robotic hardware.

• Benchmarking Suite: Establishing standardized metrics: scan duration, acoustic coverage, reconstruction fidelity, and contact force safety.

2. LITERATURE SURVEY

Prior Work & Baseline Limitations

Evaluating state-of-the-art simulators and AI frameworks to establish our technical differentiators

SonoGym (2023)

RL Gym for 2D Tracking

What They Did:
• Fast PyBullet gym environment for training 2D probe tracking and visual servoing.

Key Limitations:
• Rigid static anatomy; no soft-tissue deformation.
• Mock ultrasound slices fetched from fixed pre-rendered stacks without acoustic ray physics.
• No volumetric 3D reconstruction.

SRT & SRT-H (2024)

Surgical VLA Transformers

What They Did:
• State-of-the-art imitation learning with diffusion policies & hybrid action tokens for surgical robots.

Key Limitations:
• Evaluated purely on rigid peg-transfer and block manipulation.
• Lacks acoustic imaging feedback and continuous contact force compliance.

AMBF & SlicerROS2

ROS Middleware Bridges

What They Did:
• ROS/ROS2-based frameworks linking surgical simulators to 3D Slicer for visualization.

Key Limitations:
• Heavy multi-process serialization overhead.
• Significant IPC latency and jitter during high-frequency robot control.
• Cumbersome multi-node setup prone to message drops.

OUR APPROACH

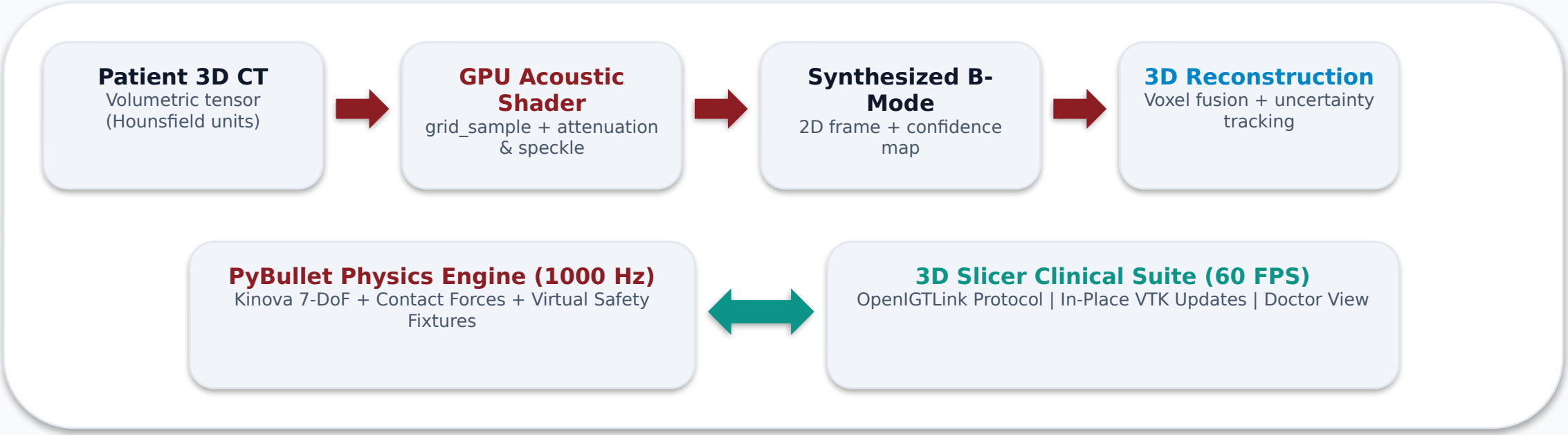
Direct, Physically Grounded

What We Do Differently:
• Direct OpenIGTLink binary link (Zero ROS overhead, sub-millisecond latency).
• Live GPU CT reslicing + physical acoustic ray tracing (attenuation, reflection, speckle).
• Soft-tissue compliance + real-time keep-out virtual safety fixtures.
• Active 3D Next-Best-View volumetric reconstruction.

3. WORK ACCOMPLISHED

System Architecture & End-to-End Workflow

Closed-loop integration of high-rate robot dynamics, GPU acoustic synthesis, and medical frontend



High-Rate Physics Loop (1000 Hz)

Simulates rigid robot dynamics, tissue deformation, joint limits, and normal force responses at 1 ms intervals to prevent numerical contact instability.

Visual & Telemetry Loop (60 Hz)

Serializes robot kinematics and contact force vectors over direct OpenIGTLink TCP sockets to 3D Slicer, updating VTK mesh transforms smoothly.

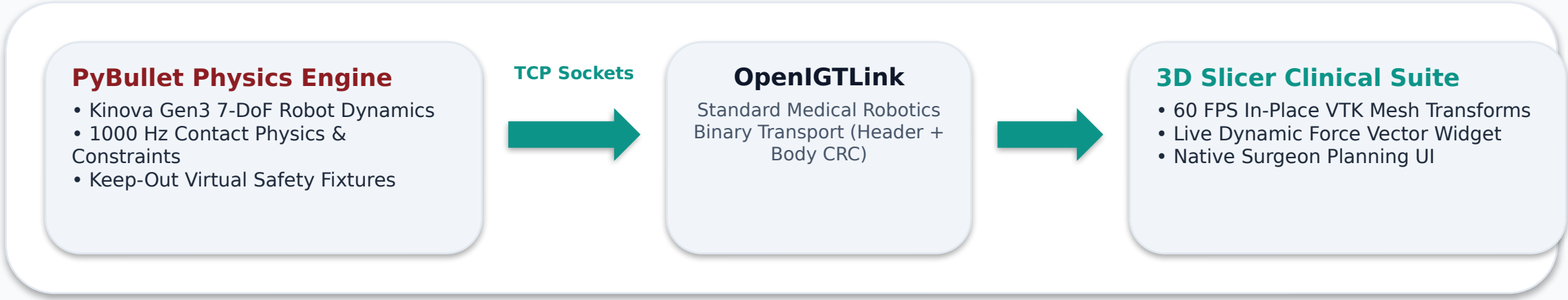
Reconstruction & Planning (10 Hz)

Accumulates synthesized 2D B-scans into a global 3D voxel grid and computes information gain to command the next-best probe trajectory.

3. WORK ACCOMPLISHED

Slicer-PyBullet Surgical Bridge

High-performance direct binary coupling between physics simulation and clinical visualization



Zero-ROS Architecture

• Why not ROS/ROS2?
ROS adds heavy serialization, daemon processes, and message drop issues on non-Linux surgical workstations.

• Direct OpenIGTLink:
Uses standard medical robotics protocol over lightweight TCP binary sockets with sub-millisecond round-trip latency.

Multi-Rate Rate Decoupling

• Stable Contact Physics:
Rigid dynamics and penalty-based contact constraints resolve at 1000 Hz (1 ms timestep), preventing probe jitter.

• In-Place VTK Updates:
3D Slicer receives state transforms and updates existing scene graph nodes in-place, eliminating memory leaks.

Virtual Safety Fixtures

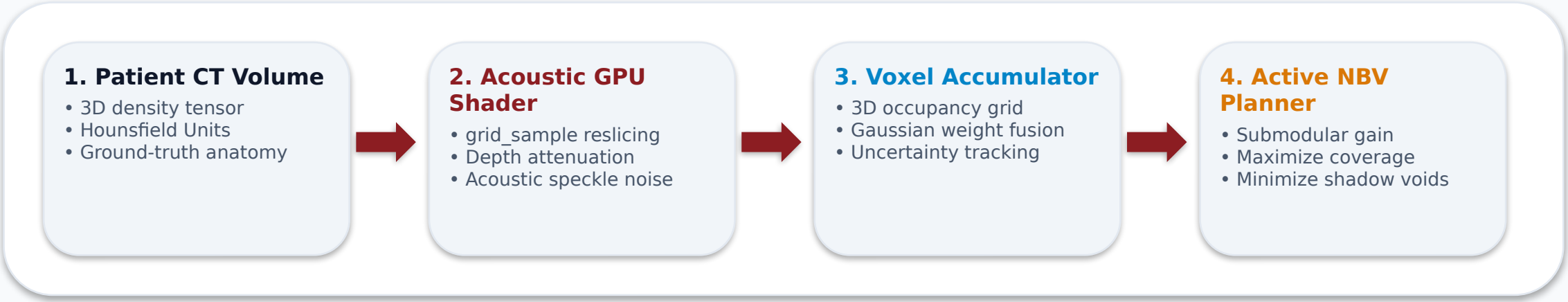
• Task-Space Keep-Out Zones:
Surgeon defines critical organ boundaries (e.g. carotid artery) in 3D Slicer; bridge streams JSON safety geometry.

• Active Force Projection:
Kinova controller projects velocity into the nullspace of boundary normals, making penetration physically impossible.

3. WORK ACCOMPLISHED

3D Ultrasound Reconstruction Pipeline

GPU-accelerated acoustic physics reslicing, confidence accumulation, and active view planning



Physical Wave Simulation

• Why not pre-rendered images?
Real ultrasound exhibits physical wave behaviors that change dynamically with probe tilt and pressure.

• GPU PyTorch Layer:
Computes depth-dependent acoustic attenuation, impedance boundary reflections, and Rayleigh speckle noise in parallel.

Voxel Confidence Fusion

• Probabilistic Fusion:
Fuses successive 2D sweeps into a continuous 3D volumetric tensor using Gaussian spatial weighting.

• Uncertainty Quantification:
Assigns lower confidence to deeper rays and acoustic shadow regions behind bone or dense tissue.

Active Next-Best-View

• Autonomous Path Planning:
Rather than blind raster scanning, the robot dynamically evaluates candidate 6-DoF probe poses.

• Information-Theoretic Gain:
Selects the next trajectory that maximizes volumetric entropy reduction while minimizing robot travel time.

4. TECHNICAL UNDERSTANDING

Architectural Decisions & Engineering Trade-Offs

Evaluating key design choices, rationale, and answers to fundamental technical questions

DECISION 1

OpenIGTLink vs. ROS2

• The Trade-Off:
ROS2 provides great tooling for general robotics, but its DDS middleware introduces significant latency, complex cross-compilation, and OS dependency locks.

• Why OpenIGTLink?
1. Native Clinical Standard: 3D Slicer natively parses OpenIGTLink binary messages without intermediate bridges.

2. Sub-Millisecond Jitter: Direct TCP binary sockets eliminate serialization bottlenecks during 1000 Hz physics loops.

DECISION 2

GPU Acoustics vs. Neural Rendering

• The Trade-Off:
Neural Radiance Fields (NeRFs) and diffusion generators produce photorealistic textures but suffer from hallucination and require hours of pre-training per patient scan.

• Why Model-Based GPU Physics?
1. Instant Generalization: Runs zero-shot on ANY new patient CT scan in milliseconds via PyTorch tensor slicing.

2. Strict Physical Validity: Acoustic wave attenuation and acoustic shadows are mathematically exact rather than hallucinated.

DECISION 3

Submodular NBV vs. Raster Sweeps

• The Trade-Off:
Fixed raster/zigzag sweeping is simple to script but wastes massive time scanning already-observed tissue while missing occluded anatomy.

• Why Submodular Information Gain?
1. Minimal Scan Duration: Prioritizes views that maximize volumetric entropy reduction.

2. Shadow Disambiguation:
Intelligently tilts the probe around acoustic obstacles (like ribs or bone) to illuminate hidden tissue.

5. FUTURE DIRECTION

Policy Learning & Sim-to-Real Roadmap

Next steps for the second half of the project: VLA integration, policy benchmarking, and phantom transfer

PHASE 1 (WEEKS 9-11)

Multimodal VLA & Policy Learning

• Action-Chunking Diffusion Policies:
Deploy Continuous Flow Matching (CFM) and diffusion action policies for smooth 6-DoF probe velocity generation.

• Multimodal Conditioning:
Condition the policy on synthesized B-mode ultrasound, contact force feedback, and natural language clinical goals (e.g., 'Reconstruct hepatic vein').

PHASE 2 (WEEKS 12-14)

Benchmarking & Evaluation Suite

• Standardized Benchmark Tasks:
Evaluate policy performance across a dataset of 20+ diverse patient abdominal and cardiac CT scans.

• Evaluation Metrics:
Measure volumetric coverage percentage, acoustic shadow volume, total scan trajectory length, and contact force boundary adherence.

PHASE 3 (WEEKS 15-16)

Sim-to-Real Hardware Transfer

• Phantom Validation:
Deploy the trained autonomous scanning policy onto a physical 7-DoF robotic manipulator scanning an ultrasound phantom.

• Domain Randomization:
Apply acoustic impedance and friction randomization during simulation training to achieve zero-shot transfer.

CONCLUSION

Project Summary & Key Takeaways

Recap of technical accomplishments, project impact, and readiness for AI benchmarking

CORE PLATFORM BUILT

Slicer Surgical Bridge

• Functional OpenIGTLink binary link connecting 1000 Hz PyBullet physics to 3D Slicer at 60 FPS.
• Real-time virtual keep-out safety fixtures preventing patient harm.

PHYSICAL WAVE SYNTHESIS

3D Reconstruction Pipeline

• GPU PyTorch reslicing with physical acoustic attenuation, impedance reflections, and speckle.
• Active submodular Next-Best-View planning to eliminate acoustic shadows.

AI BENCHMARK READY

Sim-to-Real Foundation

• Clean APIs ready for training Vision-Language-Action policies and diffusion controllers.
• A solid, reproducible foundation for final evaluation and sim-to-real transfer.

THANK YOU | QUESTIONS & DISCUSSION

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